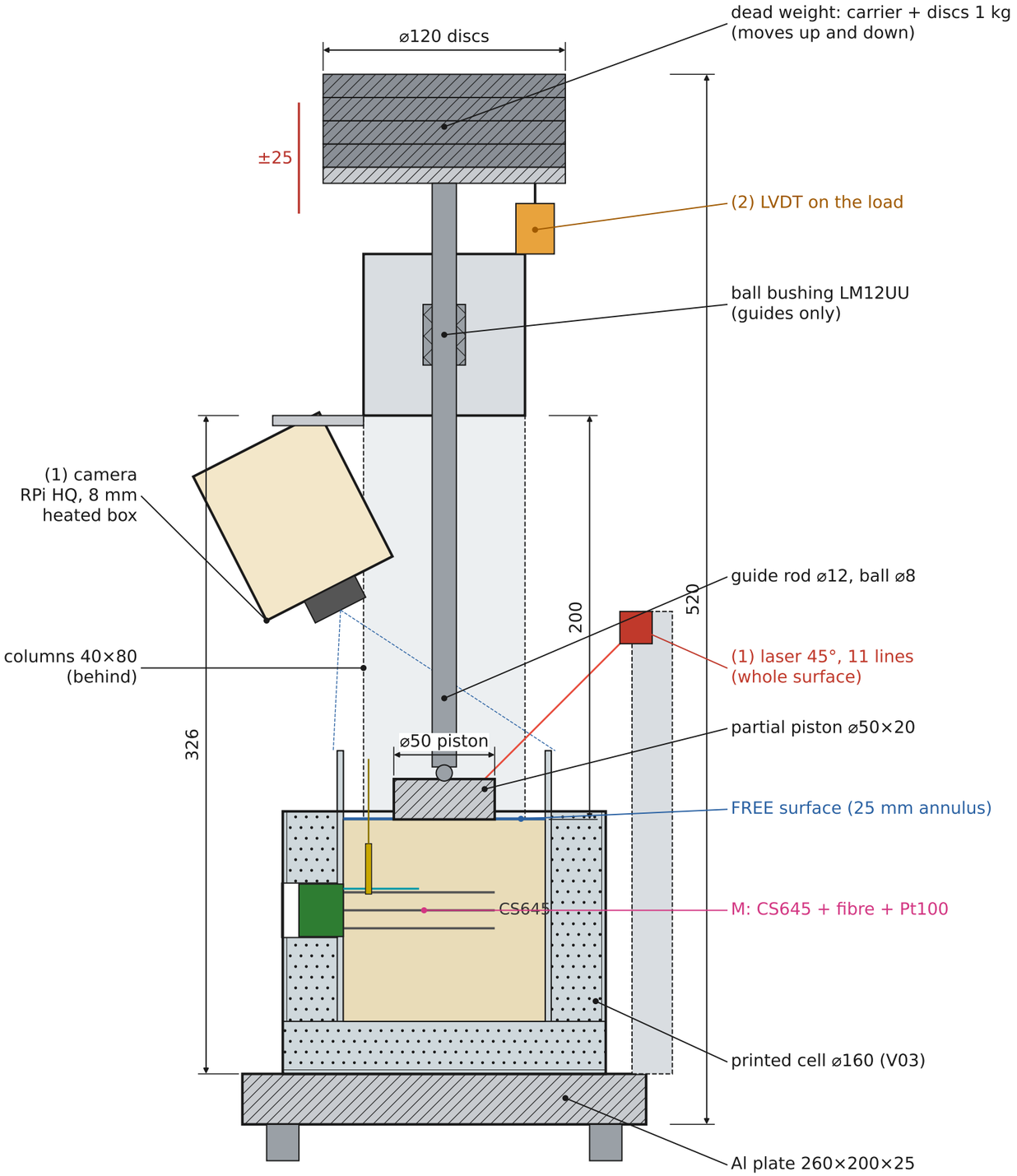


Load under the piston (ø50 = 1963 mm²)

kPa	total kg	1 kg discs	trim kg
5	1.00	0	0.39
10	2.00	1	0.39
15	3.00	2	0.39
20	4.00	3	0.39
25	5.00	4	0.39

moving part without discs ≈ 0.61 kg ≈ 3.0 kPa (minimum)

Parts

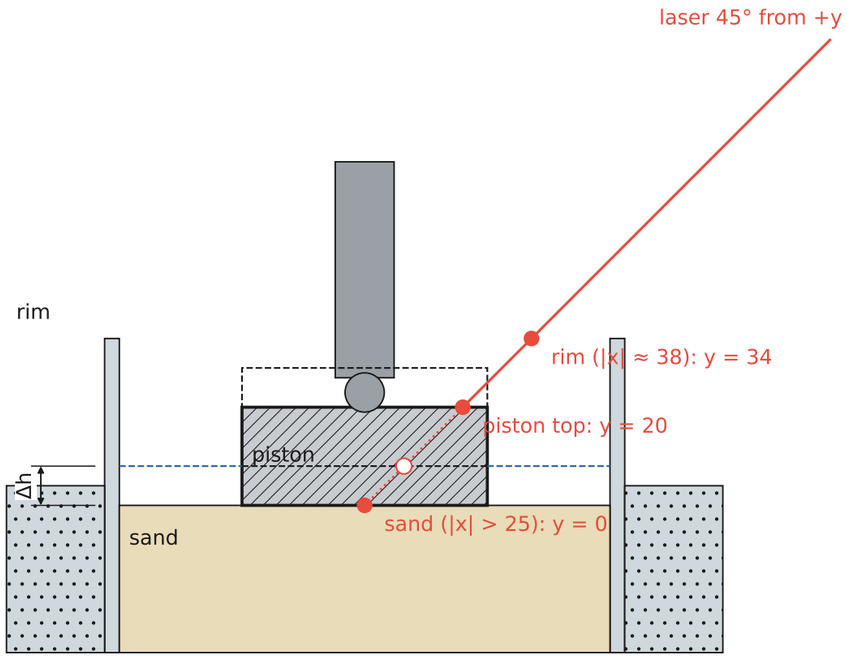
#	part
1	Al base plate 260×200×25, 3 levelling feet
2	2 columns Al 40×80 × 326, cross-beam 80×80 × 260
3	ball bushing LM12UU in the beam, rod ø12 h6
4	partial piston Al ø50×20, steel ball ø8
5	carrier Al ø120×8, steel discs ø120 (1 kg)
6	printed cell ø160 + sand ø100×100 (V03)
7	(1) camera + 11-line laser on brackets (V02)
8	(2) LVDT RDP DCTH400AG on the beam (V02)
9	point M: CS645, Pt100, optical fibre (existing)

side view y–z through the axis; the Al frame only GUIDES the load

Rig C – dead weight on a partial piston, free surface

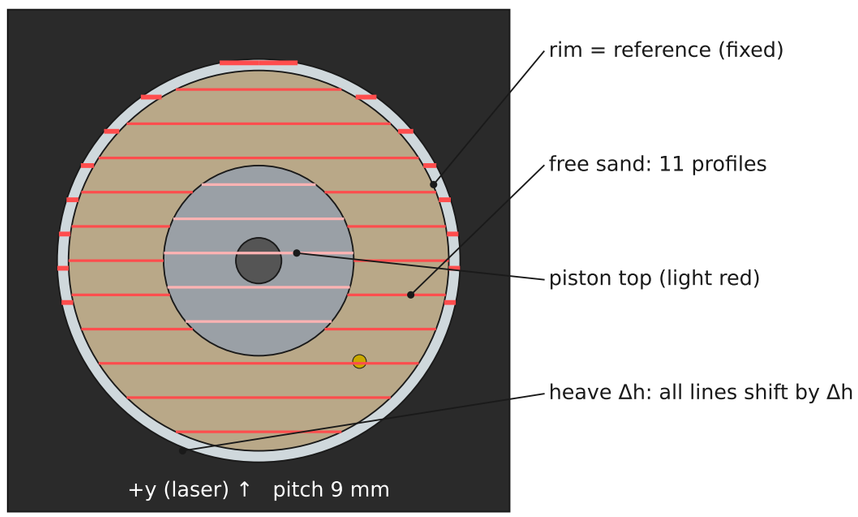
status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E7 – freeze–thaw loading tests	date 2026-10-26
scale	**≈ 1 : 2.5**	drawing no.	**E7-V01**	sheet 1 / 5

(a) LASER TRIANGULATION – section y–z 1 : 1.25

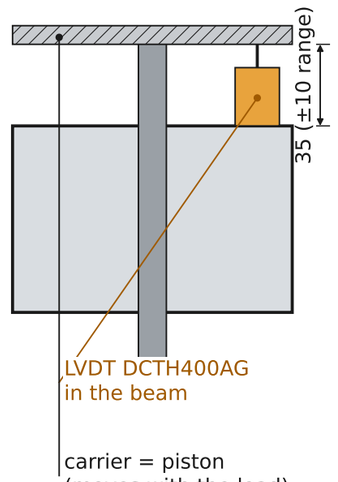


heave Δh moves the hit point by Δy = Δh (open circle)

(b) CAMERA VIEW – 11 laser lines, whole surface



(c) LVDT ON THE LOAD 1 : 2.6



What is measured

method	signal	resolution	interval
(1) camera + 11 lines	map h(x, y) of the whole surface	≈ 10 µm	every 10 / 1 min
(2) LVDT	carrier = piston	≈ 1 µm	CR1000, 10 / 1 min

Evaluation per freeze–thaw cycle

quantity	how
surface map	Δh(x, y) on 11 profiles → contour map, whole ⌀100
heave, free surface	mean Δh over the annulus (minus rim)
heave under load	Δh_load = LVDT (check: (1) on the piston)
settlement on thawing	drop from the maximum to the end of thaw
residual per cycle	height at end of thaw − start of cycle
effect of the load	Δh_free − Δh_load
strain	Δh / 100 mm
frost depth	0 °C isotherm from Pt100 / fibre / CS645 (M)

Procedure

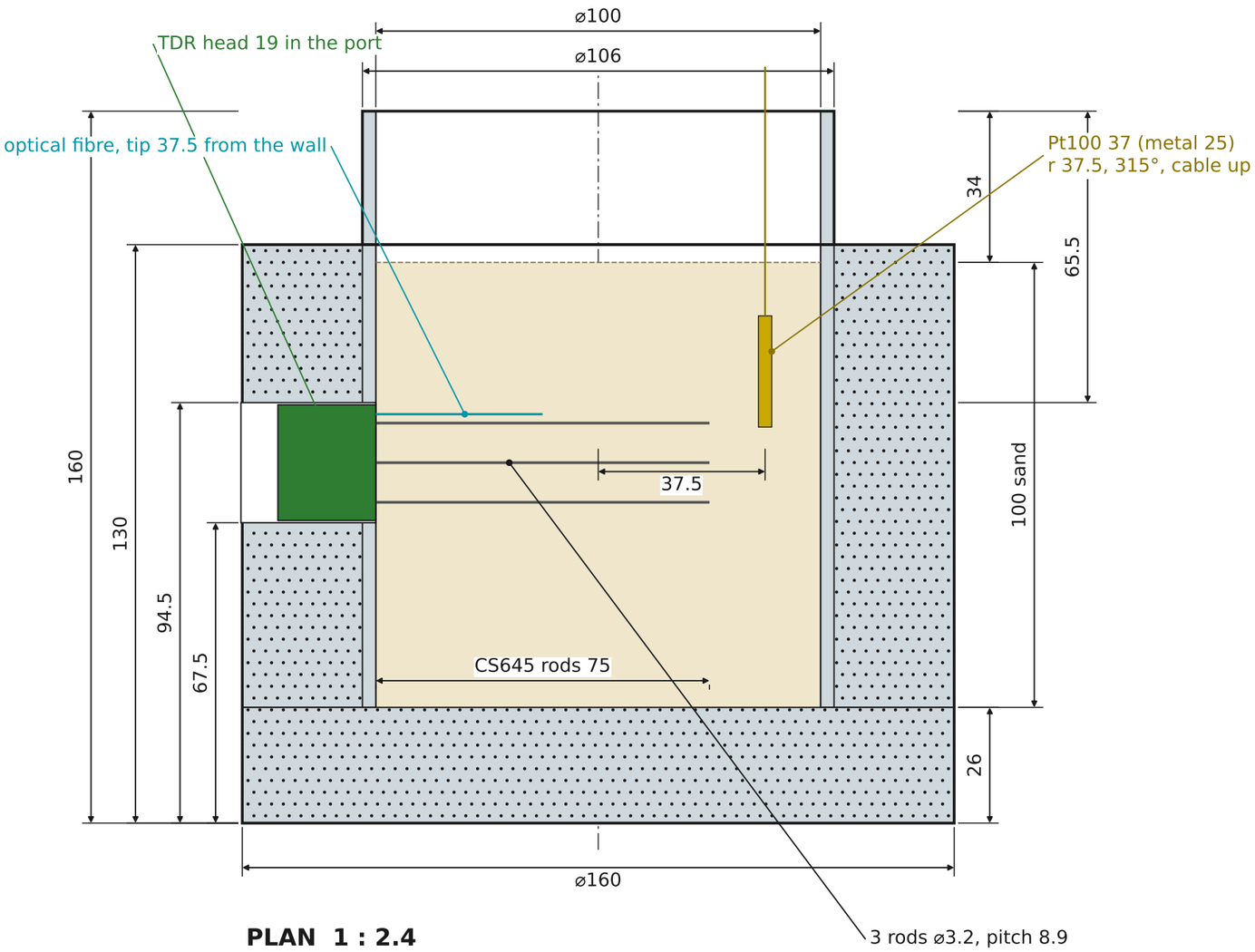
1 blank run with an Al dummy sample: thermal drift of rim, frame and LVDT (rim ≈ 120 µm / 20 K) → subtract
2 calibrate (1): lift a dummy with a micrometer in 0.5 mm steps, camera px → mm for free sand and piston top
3 load, 24 h at +10 °C, then zero (1) and (2)
4 freeze–thaw program; photo + LVDT every 10 min, 1 min around 0 °C; laser on only for the photo
5 weigh the cell before and after each cycle
6 LVDT ±10 mm: re-clamp if the heave exceeds ≈ 8 mm

heave while freezing, settlement while thawing, residual per cycle

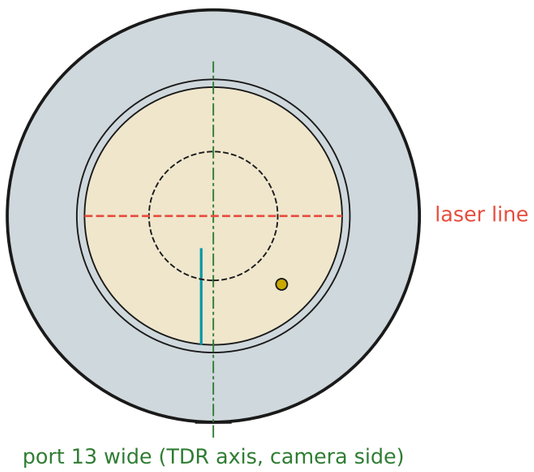
Deformation measurement – (1) camera + laser, (2) LVDT

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E7 – freeze–thaw loading tests	date 2026-10-26
scale	**various**	drawing no.	**E7-V02**	sheet 2 / 5

SECTION A–A 1 : 1.25 (along the TDR axis)



PLAN 1 : 2.4



Point M (existing sensors)

	sensor	position
TDR	Campbell CS645	3 rods 75, pitch 8.9, in the port, depth 36/45/54
OF	optical fibre	at the top rod, tip 37.5 from the wall
Pt100	37 mm (25 metal)	vertical, r 37.5 at 315° (free annulus, off the laser)

- The piston ⌀50 (dashed) covers the centre; the Pt100 sits in the free annulus on the camera side so it never crosses the laser line.
- Sand 100 mm (z 26–126), 34 mm free above (piston 20 + travel).
- Print: PETG(-CF) 0.2 mm, jacket 10 % infill = insulation; 100 % infill under the cavity, cavity wall ≈ 5 mm (≈ 8 perimeters).
- Inside: epoxy coat + PTFE spray + silicone grease.
- Cables: TDR head + fibre through the port, Pt100 straight up.
- Full dimensions: cad/E7_V04_cell.dxf.

per drawing M. Vrána 23.06.2026; z = 0 at the underside; CAD: cad/E7_V04_cell.dxf

Printed cell ⌀160 (Column_one_body_version_1) with point M

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E7 – freeze–thaw loading tests	date 2026-10-26
scale	**1 : 1.25**	drawing no.	**E7-V03**	sheet 3 / 5

To buy

tag	item	EUR	CR1000
CAM	Raspberry Pi 5 4 GB + PSU, SD, camera cable (RPishop)	105–120 *	– (Ethernet, NTP)
+	Raspberry Pi HQ Camera C/CS, IMX477 (RPishop OFI060)	50–62 *	–
+	Arducam 8 mm CS lens F1.4, manual focus (The Pi Hut)	16–20 *	–
LAS	Picotronic DOE253-635-1-6(10x22), 11 lines, 635 nm, 1 mW, 5 V	92 *	– (switched by RPi)
+	heated camera box: XPS 30, heater 5 W, double glass	40–80	–
LVDT	RDP DCTH400AG ±10 mm, spring (Elkome)	581 *	SE1, ±5000 mV
+	Mean Well HDR-15-24, 24 V DIN (floating supply) (TME)	11–14 *	via relay
+	Campbell 4WPB100 (for the existing Pt100) (Ekotechnika)	60–120	DIFF1
Tair	Campbell 109-L (optional) (Ekotechnika)	150–220	SE3, VX1

Mechanics (estimate)

item	EUR
Al base plate 260 × 200 × 25 (machined)	60–120
Al profiles 2× 40×80 × 326 + 80×80 × 260, core bolts	70–150
LM12UU bushing + rod ⌀12 h6 × 290 + steel ball ⌀8	25–60
Al partial piston ⌀50 × 20 + carrier ⌀120 × 8 (machined)	40–80
steel discs ⌀120: 4 × 1 kg + 0.5 kg + trim washers	40–90
camera bracket + laser post (profile 20×20), LVDT clamp	25–60

Total ≈ 1365–1869 EUR excl. VAT (≈ 34–47 tis. Kč). Already available: printed cell, CS645 + TDR100/200, Pt100, optical fibre, CR1000.

CR1000 channel plan

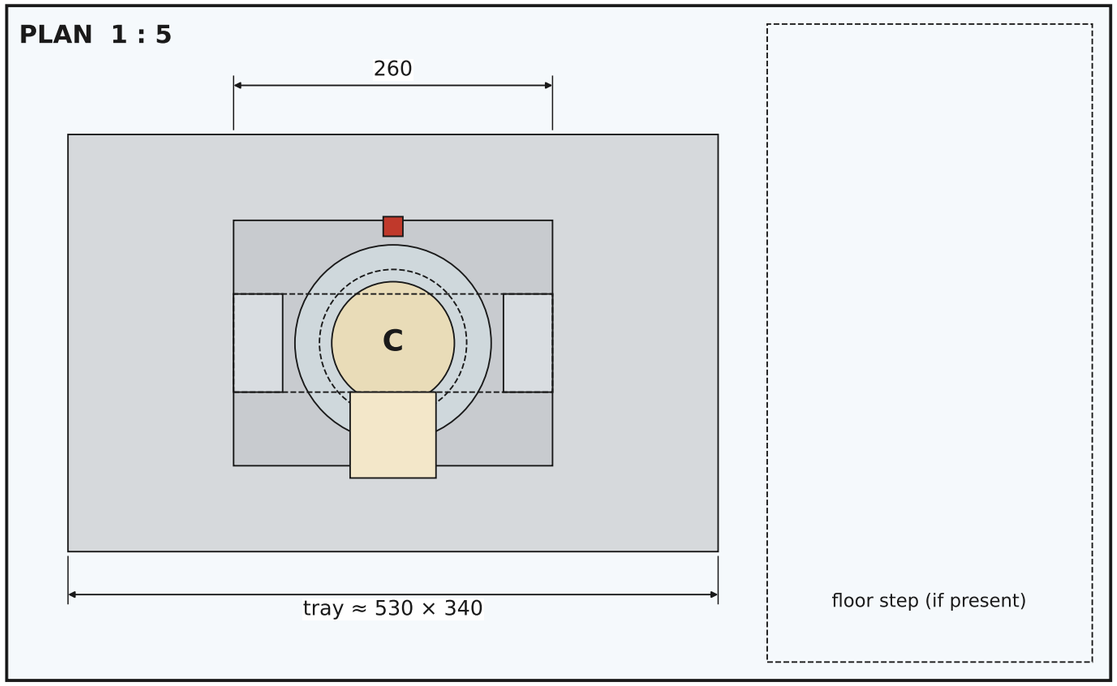
terminal	connected to	measurement
SE1	(2) LVDT	VoltSE ±5000 mV
DIFF1	Pt100 via 4WPB100	BrHalf4W, VX2 2500 mV
SE3	Tair (optional)	Therm109, VX1
C1–C3	TDR100/200 → CS645	SDM
C4	relay: LVDT supply	on for the scan
–	(1) camera + laser	Raspberry Pi, NTP
–	optical fibre	own reader

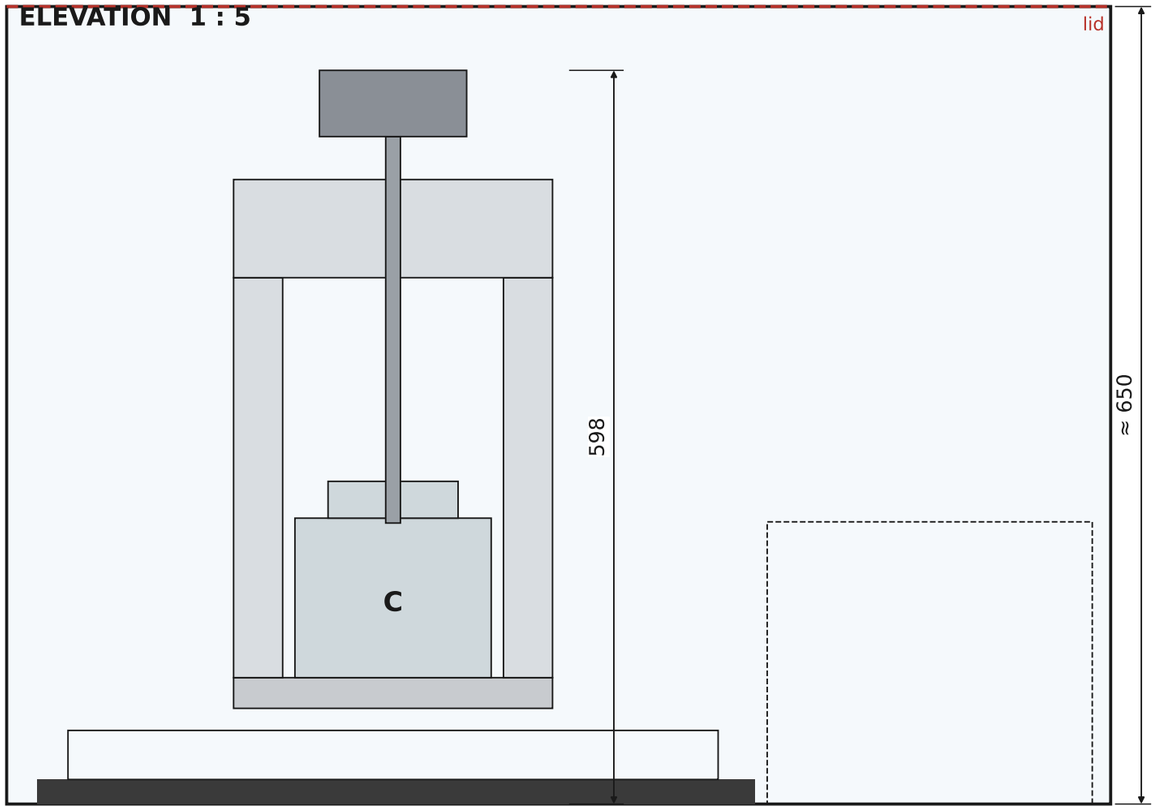
Notes
- CR1000, TDR and RPi outside the freezer; cables and the camera USB/CSI cable through the lid seal.
- Scan 10 min, 1 min around 0 °C; RPi photo at the same time stamps (CR1000 sets a flag / NTP).
- No pressure cells: the load is a known dead weight.

existing = already in the lab; prices ≈ EUR excl. VAT, * = dealer price found, rest estimated

Sensors, CR1000 wiring and costs

status	DRAFT	drawn Claude Code (draft)	checked: —
units	mm	project E7 – freeze–thaw loading tests	date 2026-10-26
scale	–	drawing no. **E7-V04**	sheet 4 / 5





Fit check

item	size	value	status
base plate	260 × 200	tray ≈ 530 × 340	OK
rig height	plate + frame + discs	538 above tray	check
highest point	above freezer floor	598	≈ 52 spare
load travel	± 25 mm above the beam	max 623	lid closes?

Notes

- The free end of the freezer stays empty (cable bundle, a second rig).
- Highest point is the disc stack; with 4 discs ≈ 50 mm under the lid – measure the freezer and the tray before ordering the profiles.
- Camera box and laser hang inside the frame – nothing outside the plate.
- Run the blank test with the Al dummy first.

same freezer as E6 (inside ≈ 900 × 550 × 650, to be measured); CR1000 + RPi outside

Rig C in the chest freezer

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E7 – freeze–thaw loading tests	date 2026-10-26
scale	**1 : 5**	drawing no.	**E7-V05**	sheet 5 / 5